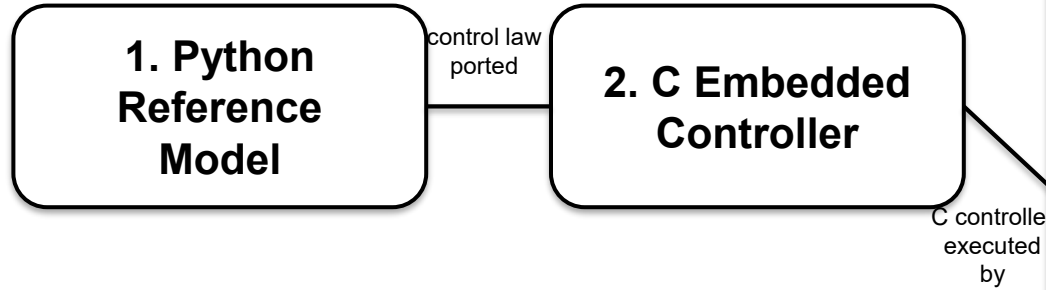
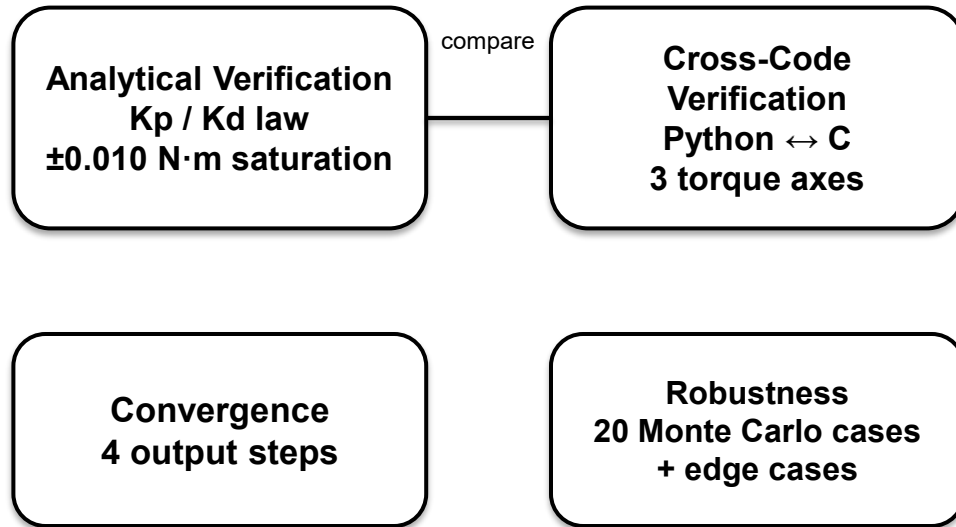


System Architecture — Real-Time GNC Loop Under Emulation

A. DEVELOPMENT / PORTING FLOW

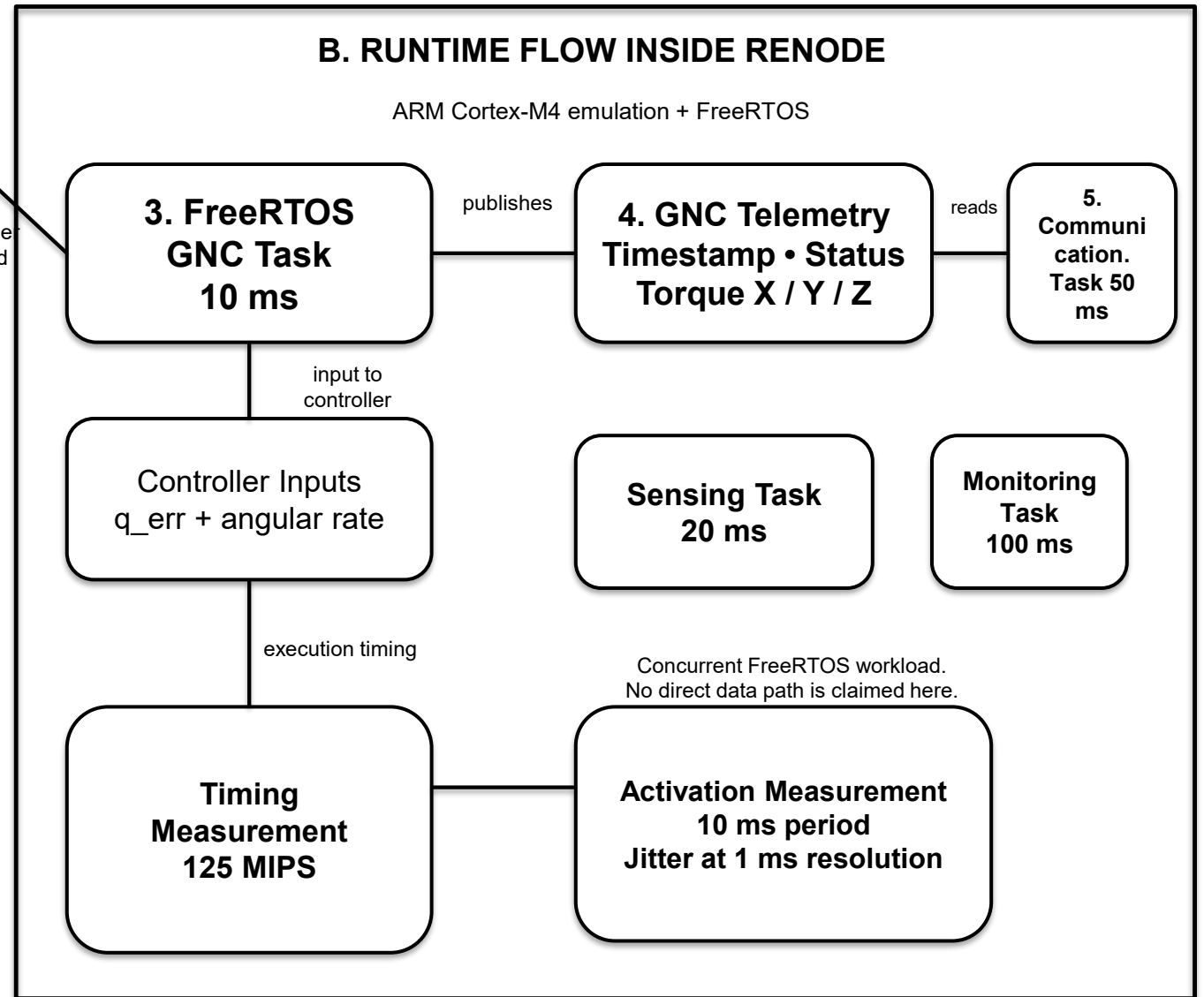


C. VERIFICATION PATHS



B. RUNTIME FLOW INSIDE RENODE

ARM Cortex-M4 emulation + FreeRTOS



MAIN RUNTIME ORDER: 3 GNC Task → 4 Telemetry → 5 Communications Task

Development order: 1 Python Reference → 2 C Controller → 3 FreeRTOS GNC Task → 4 Telemetry → 5 Communications